

C Main.c

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main() {
6     char text[100];
7     int k, i;
8
9     printf("Enter the plaintext: ");
10    fgets(text, sizeof(text), stdin);
11
12    printf("Enter key (1-25): ");
13    scanf("%d", &k);
14
15    if (k < 1 || k > 25) {
16        printf("Invalid key! Key must be between 1 and 25.\n");
17        return 0;
18    }
19
20    for (i = 0; text[i] != '\0'; i++) {
21        if (isupper(text[i]))
22            text[i] = (text[i] - 'A' + k) % 26 + 'A';
23        else if (islower(text[i]))
24            text[i] = (text[i] - 'a' + k) % 26 + 'a';
25    }
26
27    printf("Ciphertext: %s", text);
28
29    return 0;
30 }
```

Console

I/O

AI Agent

14.1s

Enter the plaintext: hi my name is sanjai

Enter key (1-25): 22

Ciphertext: de iu jwia eo owjfw